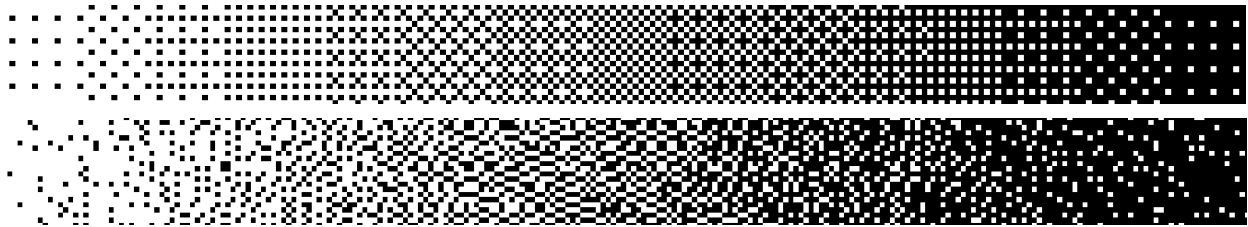


Custom Dithering Post-Processing

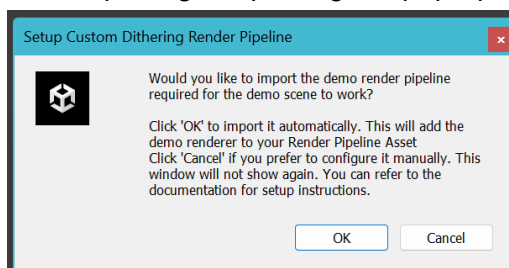


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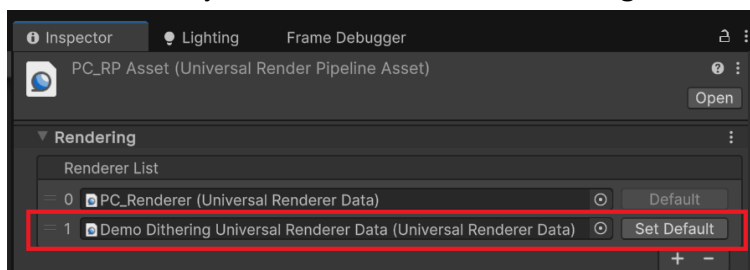
1. Setup
2. Usage
 - 2.1. Unity's Volume System
 - 2.2. Dithering Manager
3. Post-Processing Parameters
4. Demo Scenes
5. Create Custom Dithering Textures

1. Setup

- Import the package
- After importing the package, a pop-up message should appear.



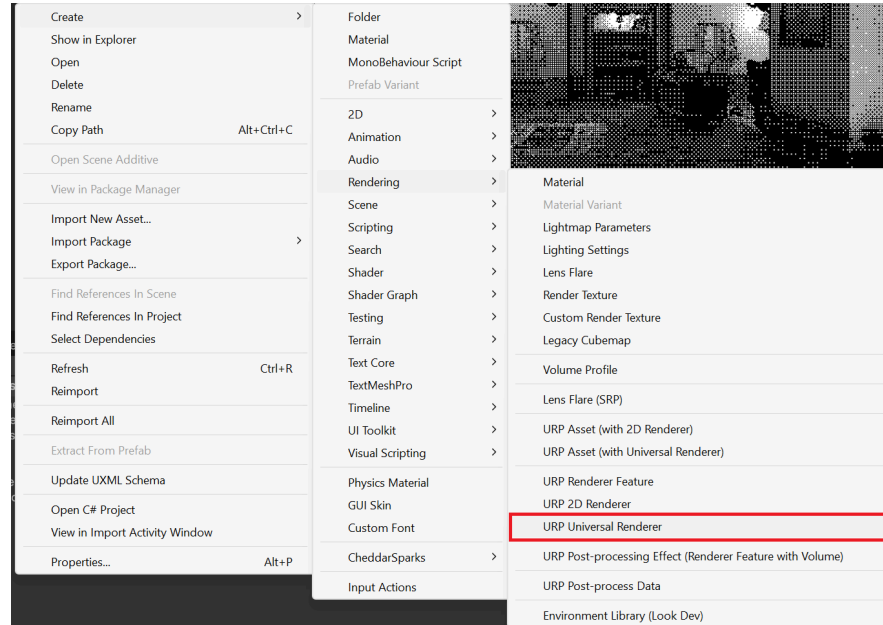
If you click **OK**, a new demo renderer will be added to your current Render Pipeline. This renderer already includes the **Custom Dithering Render Feature**.



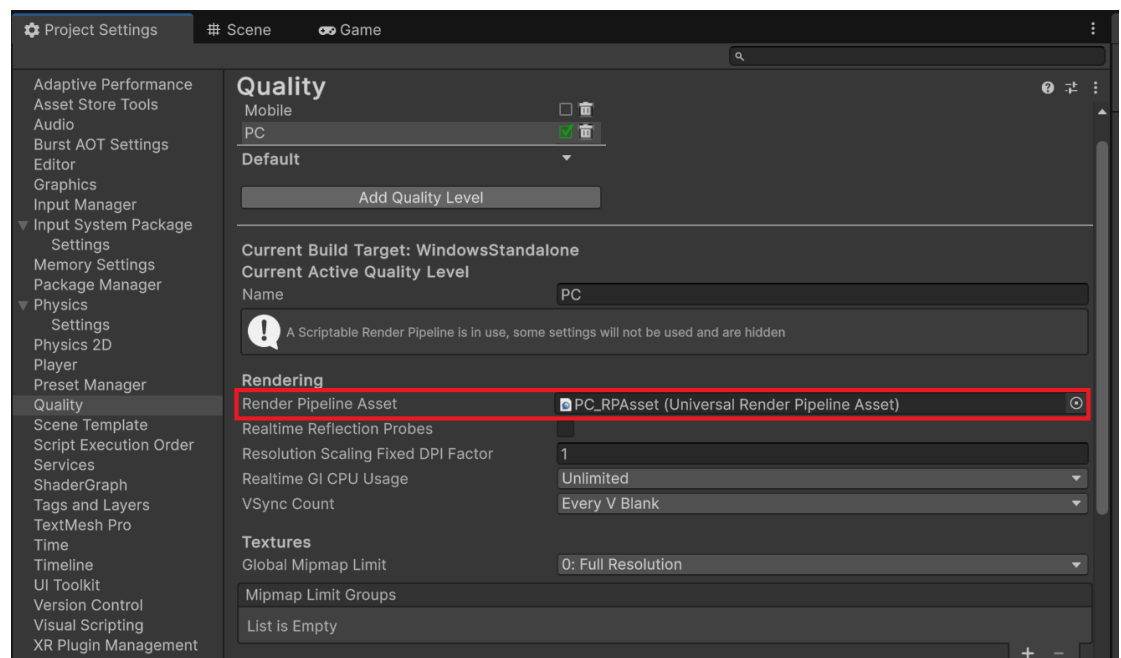
This is necessary for the demo scene to work properly.

- If you prefer to configure your Render Pipeline manually, follow these steps:
 - **[Optional]** Create a new Renderer: Right-click in the Project window > Create > Rendering > URP Universal Renderer.

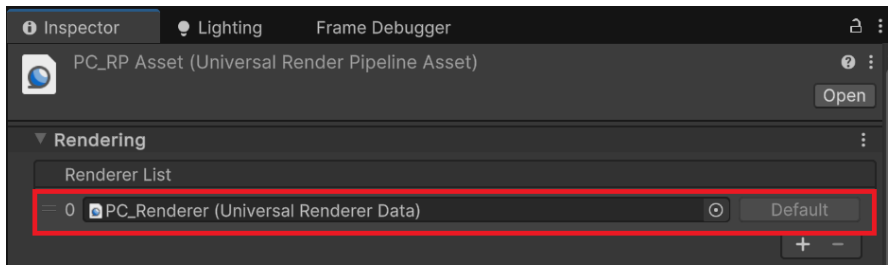
i A Renderer is a set of steps a camera uses to render a scene.



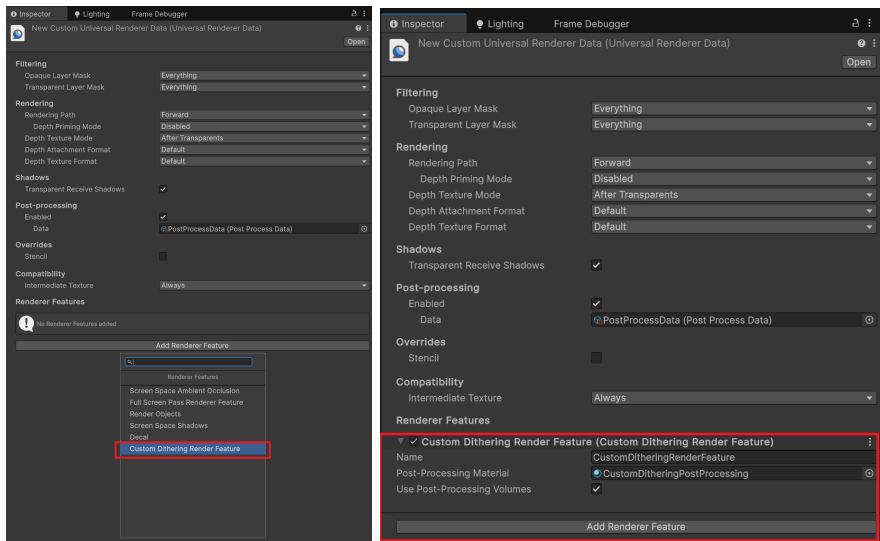
Creating a new Renderer is not necessary as you can use an existing Renderer. You can check your current Renderer in the Project Settings window > Quality > Render Pipeline Asset.



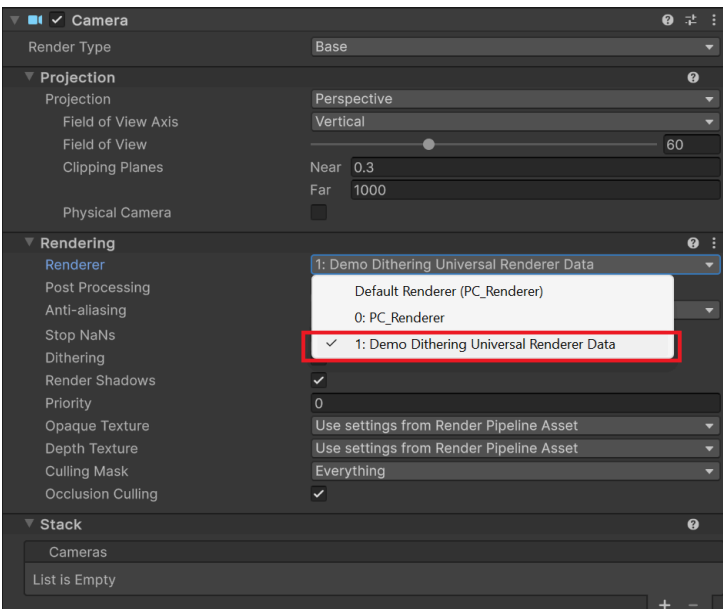
Open the **Render Pipeline Asset** of the previous image and check the **Renderer List**. Select the default **Renderer**, or add the one you just created.



- In the selected **Renderer**, add the **Custom Dithering Render Feature**.

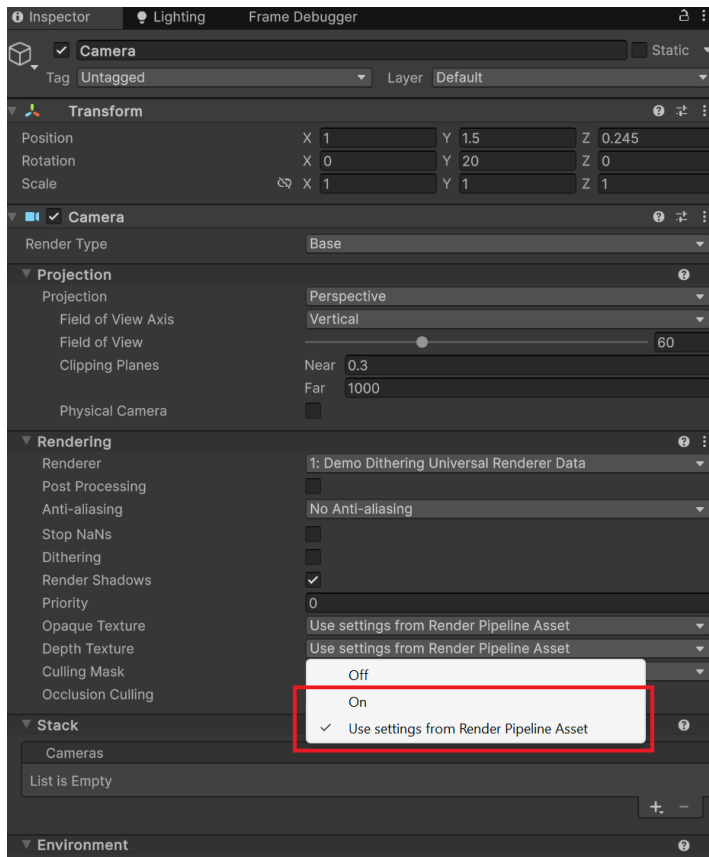


- Assign the **Renderer** to the **Camera** in your scene. (If you clicked **OK** in the initial setup message, the Demo Scene's camera is already configured)



- To use the **Depth Outline** mode, enable **Opaque Texture** in the camera settings.

***i** This is optional. Only enable it if needed, as it has a performance cost.*



2. Usage

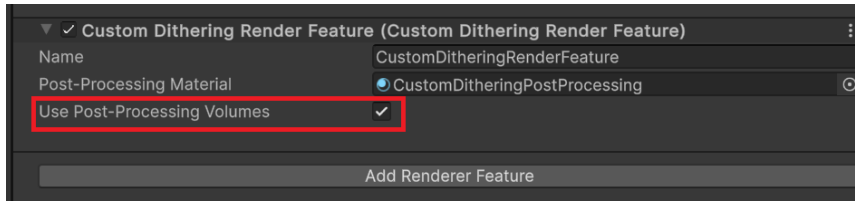
The Custom Dithering post-processing effect can be controlled via:

- Unity's **Volume System**
- The included **Dithering Manager** script

Ensure you have completed the **Setup** and that your Renderer includes the Custom Dithering Render Feature. The camera must use this Renderer.

2.1 Unity's Volume System

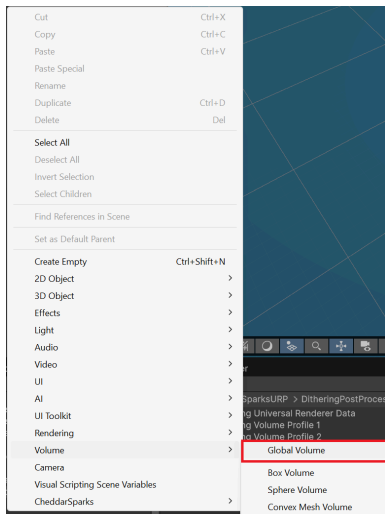
- Ensure **"Use Post-Processing Volumes"** is checked in the Render Feature settings.



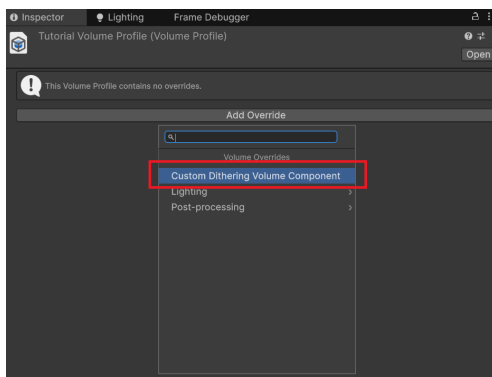
***i** The Demo Scene of the package is a perfect example of how to use the Volume system.*

- Create a Global Volume in your scene. You can create any type of volume you want.

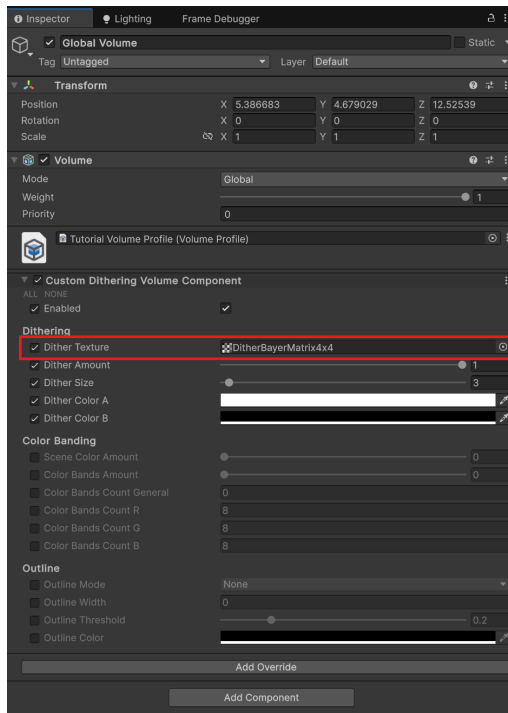
***i** Global Volumes affect all the cameras regardless of their position. Box, Sphere, and Convex Mesh Volumes only affect the cameras that are inside them.*



- **Assign a Volume Profile** to the Global Volume. You can create a new one or use some of the demo profiles provided in the package. If creating a new one, **add the "Custom Dithering Volume Component" override**.



- Configure the override. You can copy the settings from a demo profile or experiment with parameters. At minimum, **assign a Dither Texture**.



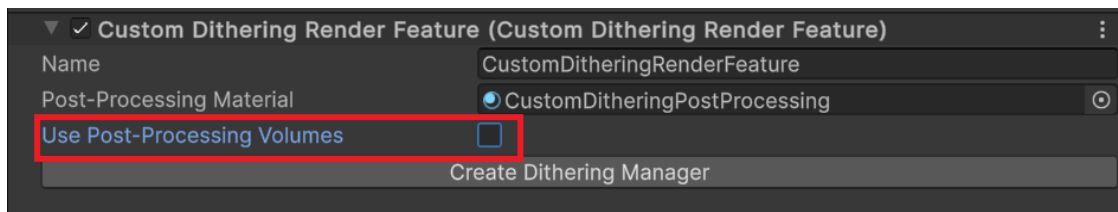
Dither Textures are found in Assets > CheddarSparksURP > DitheringPostProcessing > Textures

- At this point, you should see the effect in the **Game** window.

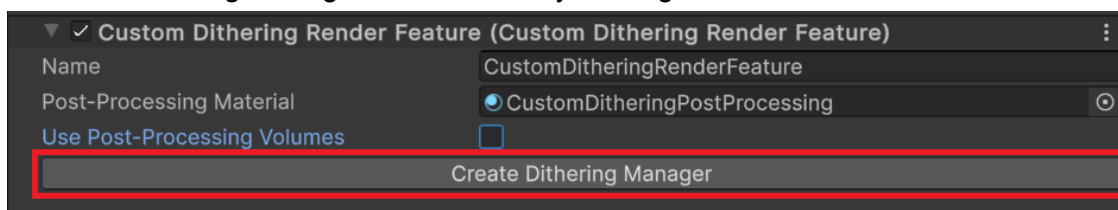
2.2 Dithering Manager

If you prefer not to use Unity's Volume System, use the **Dithering Manager** to control parameters via script.

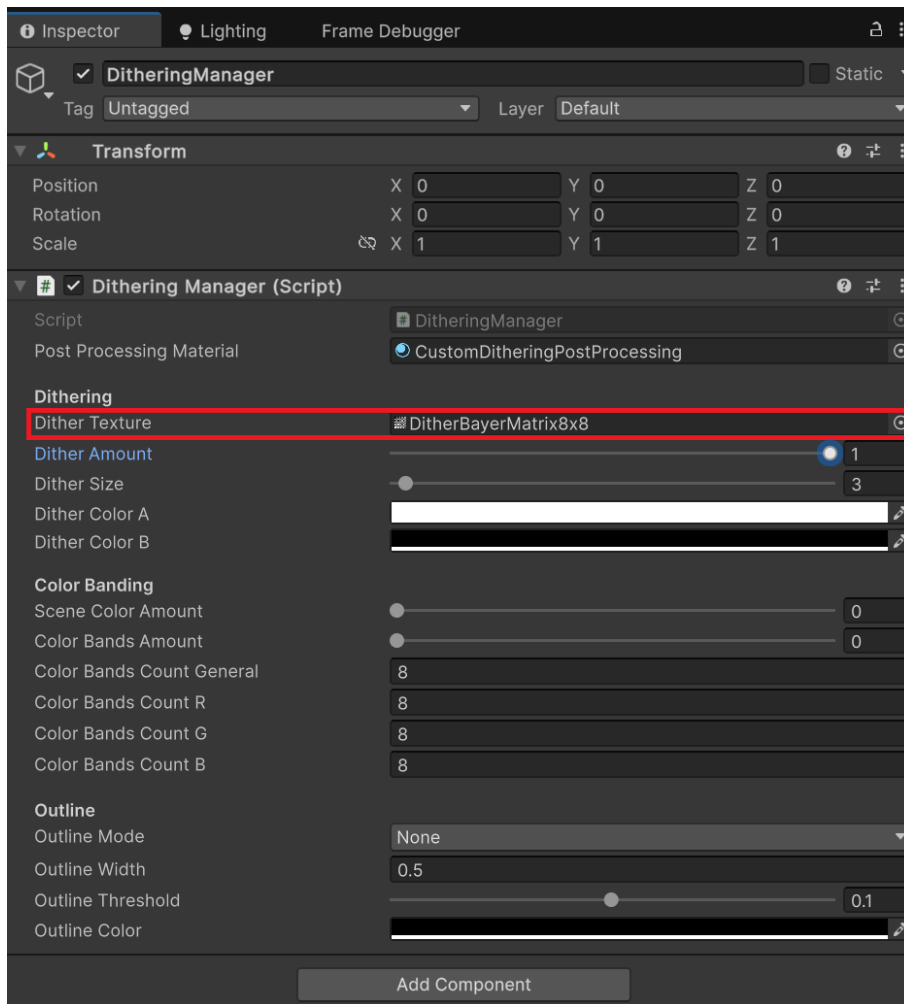
- Ensure **"Use Post-Processing Volumes"** is **unchecked** in the Render Feature settings.



- Create a Dithering Manager in the scene by clicking the button in the Render Feature.



- Assign a **Dither Texture** to the Dithering Manager.



i Dither Textures are found in Assets > CheddarSparksURP > DitheringPostProcessing > Textures

- Adjust parameters in the Inspector and observe changes in the **Game** window.
- The **DitheringManager** component can also be accessed via script to update parameters at runtime:

```
void SetPostProcessingMaterial(Material material)
void SetDitherTexture(Texture2D ditherTexture)
void SetDitherAmount(float ditherAmount)
void SetDitherSize(int ditherSize)
void SetDitherColorA(Color ditherColorA)
void SetDitherColorB(Color ditherColorB)
void SetSceneColorAmount(float sceneColorAmount)
void SetColorBandsAmount(float colorBandsAmount)
void SetColorBandsGeneral(float colorBandsGeneral)
void SetColorBandsCountR(float colorBandsCountR)
```

```

void SetColorBandsCountG(float colorBandsCountG)
void SetColorBandsCountB(float colorBandsCountB)
void SetOutlineMode(OutlineMode outlineMode)
void SetOutlineWidth(float outlineWidth)
void SetOutlineThreshold(float outlineThreshold)
void SetOutlineColor(Color outlineColor)

```

3. Post-Processing Parameters

These parameters are available in both the Volume Component and the Dithering Manager.

 *The Demo Scene offers examples of how to use each.*

- **Dither Texture:** Black-and-white texture that creates the dithering pattern.
- **Dither Amount:** Controls the strength of the dithering pattern.
 - When **Scene Color Amount** = 0 → interpolates between **Dither Color A** and the dithering pattern
 - When **Scene Color Amount** = 1 → applies pattern over the scene color (higher values may look bad)
- **Dither Size:** Controls the size of the dithering pixels.
- **Dither Color A:** Only works when the parameter *Scene Color Amount* is set to 0. It is the color of the dithering pixel of the brightest colors.
- **Dither Color B:** Only works when the parameter *Scene Color Amount* is set to 0. It is the color of the dithering pixel of the darkest colors.
- **Scene Color Amount:** Controls how much of the original scene color is used in the final effect.

 *For best results: use small values for **Dither Amount** and enable **Color Bands**. See: **Demo Dithering Volume Profile 3***

- **Color Bands Amount:** Only works when the parameter *Scene Color Amount* is set to 1. Controls how strongly the color quantization is applied.
- **Color Bands Count General:** Utility parameter that applies the same value to *Color Bands Count R*, *Color Bands Count G* and *Color Bands Count B*. Sets the number of color bands for all color channels. Higher values will show smoother color gradients and vice versa.
- **Color Bands Count R, G and B.** How many color bands will be calculated per color channel. Higher values will show smoother color gradients and vice versa.
- **Outline Mode:**  May affect performance. Choose from:
 - **Sobel:** [A sobel filter](#) will be applied to the scene luminance. It will show outlines in texture color changes and different geometries.
 - **Depth:** It will use the camera depth texture to create outlines on the edges of geometry (it does not detect edges on textures). Ensure that the depth texture is enabled in the render pipeline settings or camera settings
 - **None:** Disables outlines (no performance impact).
- **Outline Width:** Controls the thickness of the outline.
- **Outline Threshold:** Sensitivity to edge detection. Lower values = more outlines.

- **Outline Color:** Sets the color of the outlines.
 - If Scene Color Amount = 0 → outline is converted to grayscale
 - If Scene Color Amount > 0 → outline is tinted

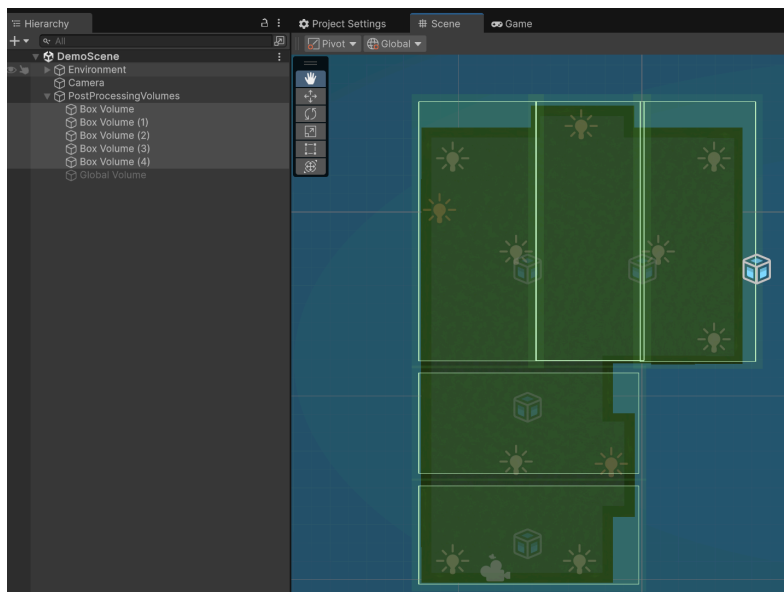
 **The Alpha value controls the transparency/intensity of the outline.**

4. Demo Scenes

Two demo scenes are included in the package to learn how to use the volume profiles and the post-processing parameters.

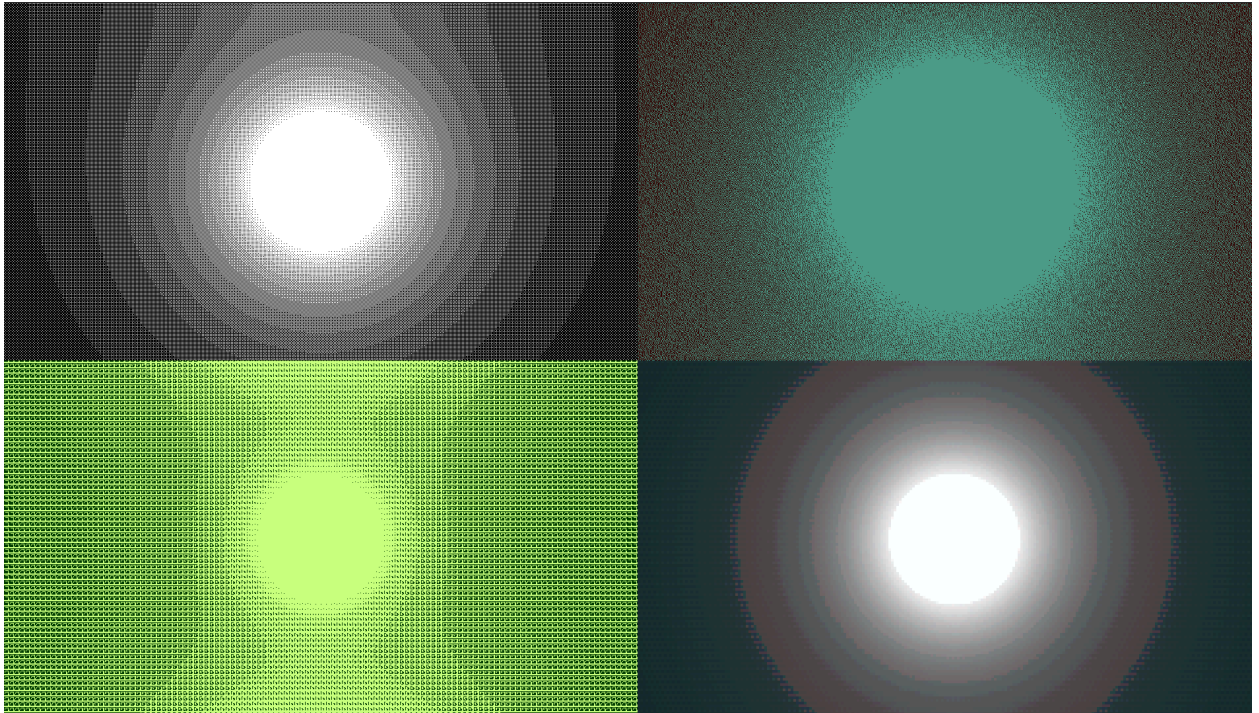
- **Demo Scene:** A furnished house with multiple post-processing volumes. You can fly around the house to see how the post-processing is applied to the camera and how it is interpolated between each volume.

The volumes are inside the game object *PostProcessingVolumes* (Green boxes in the picture):



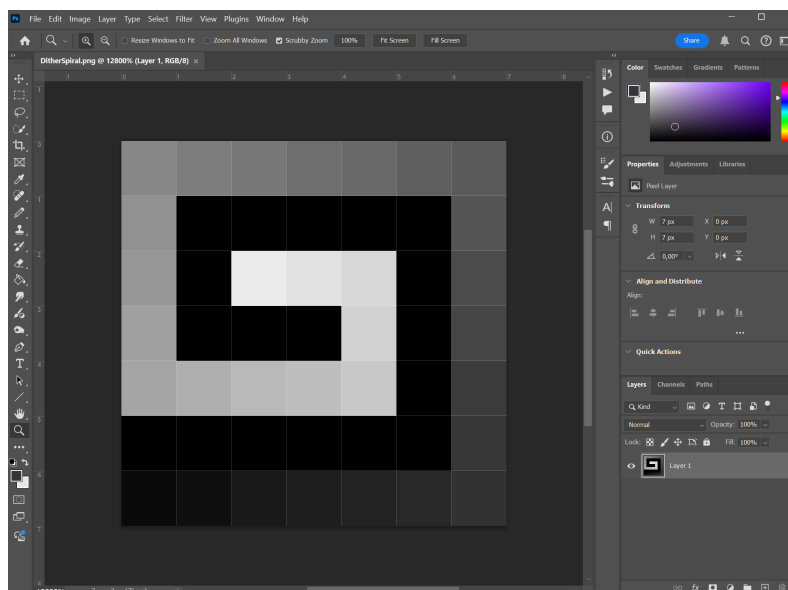
Click Play on the scene and fly around it using the same controls as in Unity's Scene window. (**WASD** to move, **Q/E** to move vertically, **Right-click + Mouse drag** to look around).

- **Demo Gradients:** A simple gradient testing scene with multiple cameras to demonstrate how dithering interacts with light gradients.



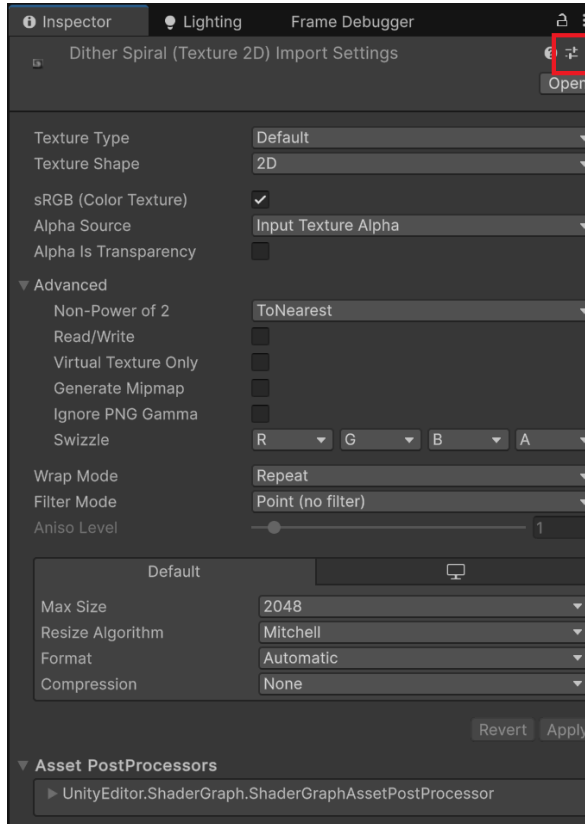
5. Create Custom Dithering Textures

You can create your own dithering textures using any image editing software. Ensure the texture uses **grayscale values (black & white only)**.

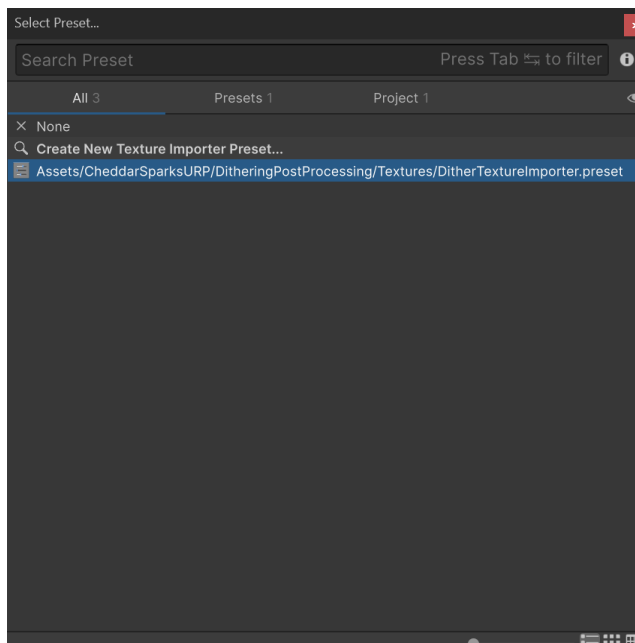


After creating your texture follow these steps to use it in the post-processing effect:

- Save your texture as **.PNG** and import it into the Unity project.
- Adjust the texture settings using the included Texture Importer preset to ensure the texture is correctly configured for the dithering effect.
- Click on this icon in the Inspector view of the texture.



- Select the DitherTextureImporter preset



- Click **Apply** to finalize the import settings.

